

Chia-Chien (Aaron) Hung

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EDUCATION

Georgia Institute of Technology

Atlanta, GA

Bachelor of Science in Computer Science, Concentration in Intelligence and Modeling/Simulation August 2023 - Present

- GPA: 4.0/4.0 (Faculty Honors)
- Relevant Coursework: Machine Learning, Artificial Intelligence, Design and Analysis of Algorithms, Computer Organization and Programming, Systems and Networks, Differential Equations, Data Structures and Algorithms

EXPERIENCE

Software Engineering Lead

Atlanta, GA

Molecular Evolution Core Lab, Georgia Tech

January 2024 - Present

- Led 15-member team developing 5 modular Lab Information Management Systems used by active research labs.
- Architected backend (Express.js + TypeScript) and frontend (React/Vite/Tailwind) with custom RBAC and GT SSO, providing HIPAA-grade data security.
- Built CI/CD infrastructure (GitHub Actions + Docker Compose) and private npm registry, halving feature deployment time and enabling automated testing.
- Refactored legacy code and migrated databases, cutting maintenance time 8×, eliminating major crashes, and reducing client data transfer time 10× via SFTP integration.
- Authored complete developer documentation and onboarding pipelines, eliminating recurring production failures.

Undergraduate Teaching Assistant

Atlanta, GA

CS 2200: Systems and Networks

August 2025 - Present

- Designed and built grading automation tools with Selenium, cutting TA workload ~70% and turnaround time.
- Collecting usage analytics for funding reports impacting ~200 teaching assistants for 5+ courses.
- Taught processor implementation, pipeline optimization techniques, and cache coherence to a class of 40+ students.

Student Researcher

Atlanta, GA

Integrative Systems Biology Lab, Georgia Tech

December 2024 - Present

- Developed C++ reaction-diffusion models of dynein-driven metastasis using Eigen and OpenMP for parallel simulation.
- Optimized $O(n^2)$ cell-update routines to $O(n)$ using sparse matrices and coefficient precomputation, achieving ~40× speed-up on Slurm HPC with larger models run for longer time periods.
- Building this work into undergraduate thesis to demonstrate computational predictions prior to wet-lab experiments.

Student Researcher

Atlanta, GA

SIPLab, Georgia Tech

May 2024 - August 2025

- Implemented neural network model predictive controller (NNMPC) in PyTorch for nonlinear Lorenz system dynamics.
- Rewrote controller in C++ and using OSQP, achieving 25× speed-up with vectorization and low-level optimization.
- Automated training and simulation pipelines with Python and Snakemake to streamline large-scale spiking neural network experiments consisting of 80 models and 200 experiments.

PROJECTS

Doc McQuery — AI Healthcare Research Platform

Atlanta, GA

HackGT 12 • 2nd Place, Impiricus Track

- Built React + Flask web app using PostgreSQL (pgvector) and Sentence Transformers for semantic search of PubMed and EMR datasets, reducing time-to-insight by ~50% for healthcare researchers.
- Integrated LLM-based summarization to generate real-time adaptive search queries and literature summaries.
- Designed system architecture, led backend infrastructure (Docker setup, vector embedding), and developed frontend.

SKILLS AND COMPETENCIES

Programming Languages: C/C++, Python, TypeScript/JavaScript, Java, HTML/CSS, LaTeX, Assembly

Frameworks & Tools: React, Express.js, Flask, Git, Docker, CI/CD, Linux, CMake, Neovim

Scientific & HPC: Eigen, OpenMP, PyTorch, Brian2, Snakemake, Numerical Optimization

Specialties: Backend Architecture, High-Performance Computing, Numerical Simulation, Systems Programming

Competitive Programming: Codeforces Rating: 1797